



Wireshark Lab: HTTP

Having gotten our feet wet with the Wireshark packet sniffer in Lab 1, we're now ready to use Wireshark to investigate protocols in operation. In this lab, we'll explore several aspects of the HTTP protocol: the basic GET/reply interaction, HTTP message formats, retrieving large HTML files, retrieving HTML files with embedded URLs, persistent and non-persistent connections, and HTTP authentication and security.

As is the case with all Wireshark labs, the full description of this lab is available at this book's Web site, <http://www.awl.com/kurose-ross>.



VideoNote
Using Wireshark to
investigate the
HTTP protocol



Wireshark Lab: DNS

In this lab, we take a closer look at the client side of the DNS, the protocol that translates Internet hostnames to IP addresses. Recall from Section 2.5 that the client's role in the DNS is relatively simple—a client sends a query to its local DNS server and receives a response back. Much can go on under the covers, invisible to the DNS clients, as the hierarchical DNS servers communicate with each other to either recursively or iteratively resolve the client's DNS query. From the DNS client's standpoint, however, the protocol is quite simple—a query is formulated to the local DNS server and a response is received from that server. We observe DNS in action in this lab.

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Marc Andreessen

Marc Andreessen is the co-creator of Mosaic, the Web browser that popularized the World Wide Web in 1993. Mosaic had a clean, easily understood interface and was the first browser to display images in-line with text. In 1994, Marc Andreessen and Jim Clark founded Netscape, whose browser was by far the most popular browser through the mid-1990s. Netscape also developed the Secure Sockets Layer (SSL) protocol and many Internet server products, including mail servers and SSL-based Web servers. He is now a co-founder and general partner of venture capital firm Andreessen Horowitz, overseeing portfolio development with holdings that include Facebook, Foursquare, Groupon, Jawbone, Twitter, and Zynga. He serves on numerous boards, including Bump, eBay, Glam Media, Facebook, and Hewlett-Packard. He holds a BS in Computer Science from the University of Illinois at Urbana-Champaign.



How did you become interested in computing? Did you always know that you wanted to work in information technology?

The video game and personal computing revolutions hit right when I was growing up—personal computing was the new technology frontier in the late 70's and early 80's. And it wasn't just Apple and the IBM PC, but hundreds of new companies like Commodore and Atari as well. I taught myself to program out of a book called "Instant Freeze-Dried BASIC" at age 10, and got my first computer (a TRS-80 Color Computer—look it up!) at age 12.

Please describe one or two of the most exciting projects you have worked on during your career. What were the biggest challenges?

Undoubtedly the most exciting project was the original Mosaic web browser in '92-'93—and the biggest challenge was getting anyone to take it seriously back then. At the time, everyone thought the interactive future would be delivered as "interactive television" by huge companies, not as the Internet by startups.

What excites you about the future of networking and the Internet? What are your biggest concerns?

The most exciting thing is the huge unexplored frontier of applications and services that programmers and entrepreneurs are able to explore—the Internet has unleashed creativity at